

(21)

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12 20	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	LOKESH S	Reg. No.:	192572151
Section / Batch:	CSE (AI)	Date:	16/07/26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

- A computer manufacturer is designing a workstation for engineering simulations that frequently exchanges data between the processor, memory, and peripheral devices. Design a concept map illustrating the interaction among the CPU, Memory Unit, Input Unit, Output Unit, and I/O devices, showing how instructions and data move through the system during program execution. How do the functional units of a computer system cooperate to execute instructions efficiently?
- A software developer notices that an application takes longer to execute because several instructions require multiple memory accesses. Develop a concept map showing how the Fetch, Decode, and Execute stages of the instruction cycle are related to memory access, instruction execution, and control flow. How does each stage of the instruction execution cycle contribute to the successful execution of a program?
- A data center experiences communication delays because the processor, memory, and I/O devices frequently compete for access to the system bus. Prepare a concept map demonstrating the relationships among Single-bus, Multi-bus, and Hierarchical Bus organizations and their connections to the CPU, Memory, and I/O devices. How do different bus organizations improve communication efficiency and reduce system bottlenecks?
- A microprocessor programmer is developing software for both 8085 and 8086 systems and must understand how instructions access data stored in memory. Represent the relationships among the instruction sets, register organization, addressing modes, memory organization, and memory segmentation (8086) using a concept map. How do the architectural differences between the 8085 and 8086 affect instruction execution and memory management?
- A firmware engineer is examining machine code stored in memory while debugging an embedded application. Create a concept map explaining the relationships between Binary and Hexadecimal Number Systems, instruction representation, memory storage, data conversion, and CPU interpretation. How do binary and hexadecimal representations simplify instruction execution, memory organization, and debugging in computer systems?

Code of Conduct

I S. Lokesh (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Lokesh

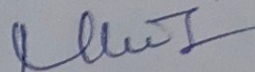
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

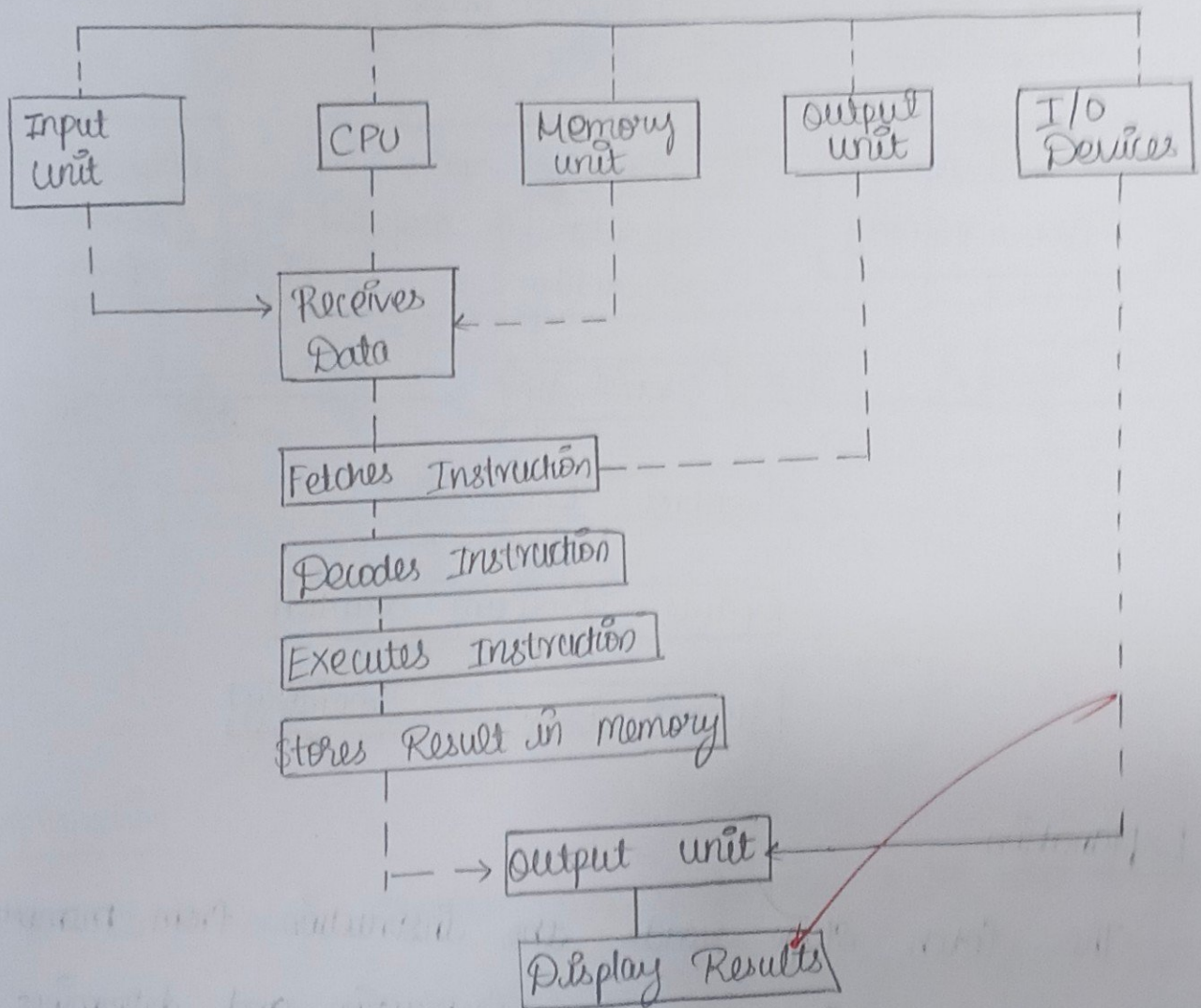
Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	4 / 4	4 / 4	3 / 4	3 / 4	3 / 4	
2	4 / 4	4 / 4	3 / 4	3 / 4	3 / 4	
3	4 / 4	3 / 4	4 / 4	3 / 4	3 / 4	
4	3 / 4	3 / 4	3 / 4	3 / 4	4 / 4	
5	3 / 4	3 / 4	3 / 4	3 / 4	4 / 4	
	23 / 25	21 / 25	16 / 20	15 / 20	9 / 10	84 / 100

Course Coordinator


Faculty Incharge

① concept map of functional units of a computer system

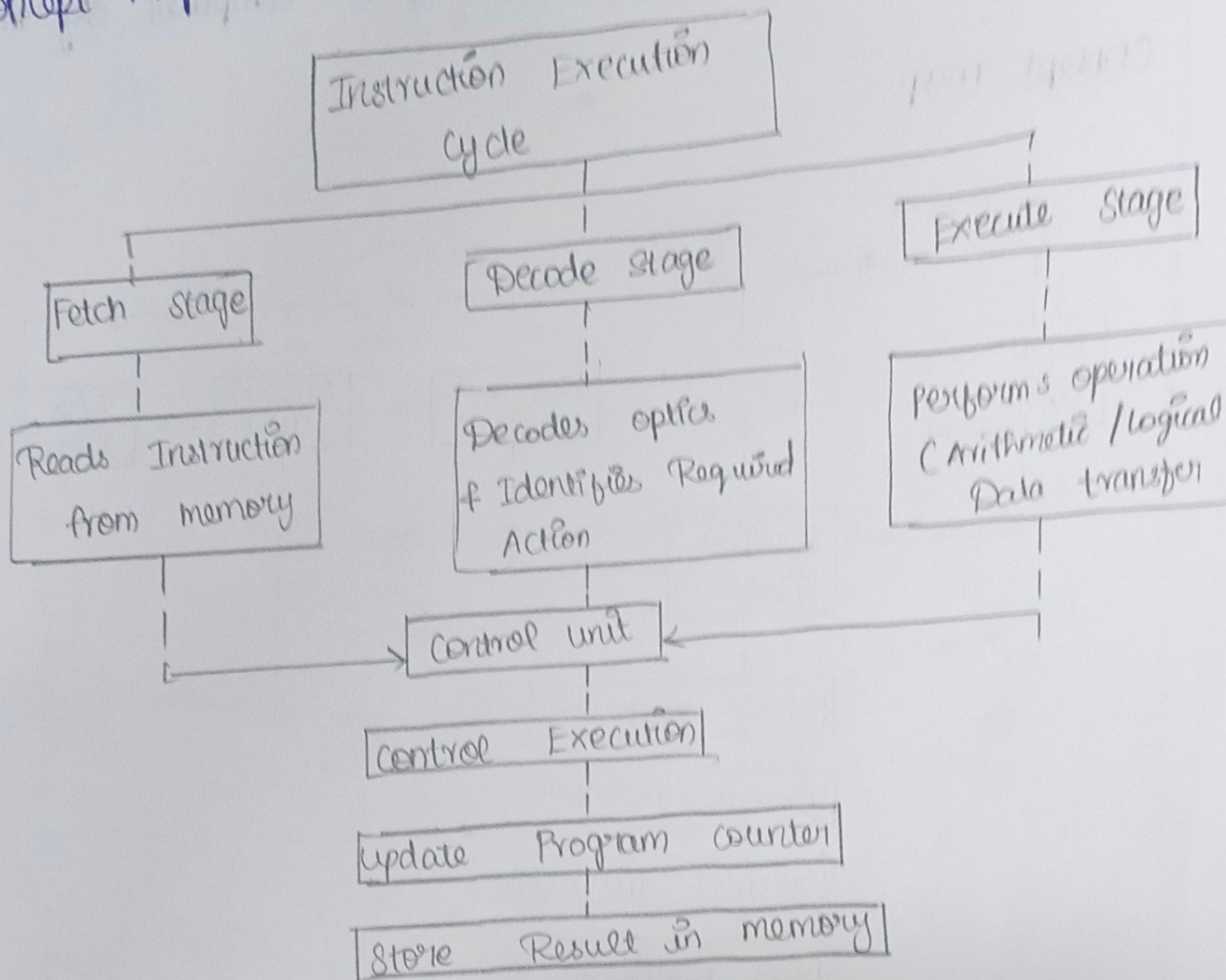
Concept map:



Explanation

The input unit receives data and sends it to CPU. The CPU fetches instructions from the memory unit, decodes them and executes them. The processed data is stored in memory and sent to the output unit through the system bus. I/O devices exchange data between the

2) Concept map of Instruction Execute Cycle.

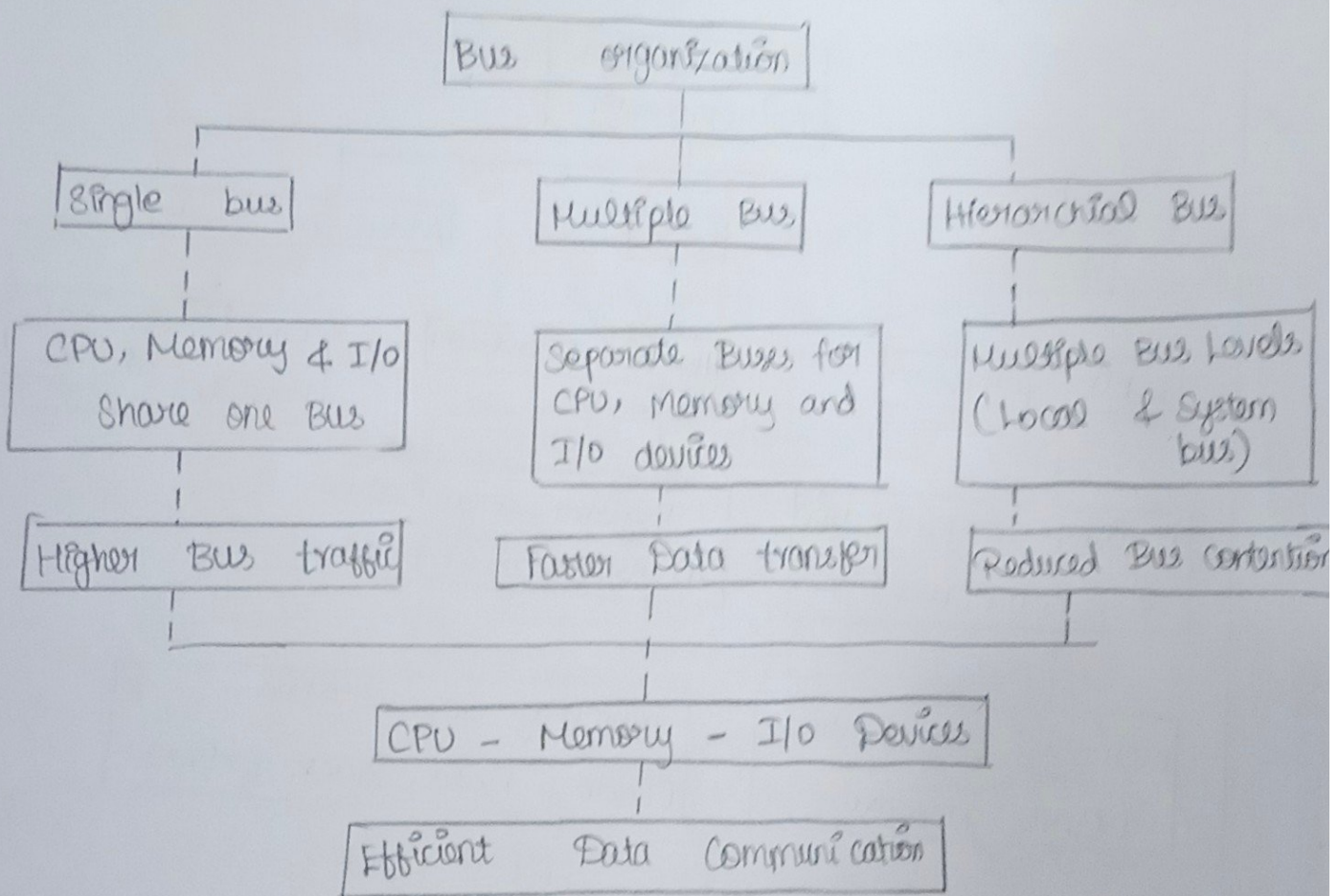


Explanation

The fetch stage reads the instruction from memory. The decode stage interprets the instruction and determines the required operation. The execute stage performs operation, updates the program counter, and stores the result in memory.

3) concept map Bus organization in Computer Architecture

Concept map:

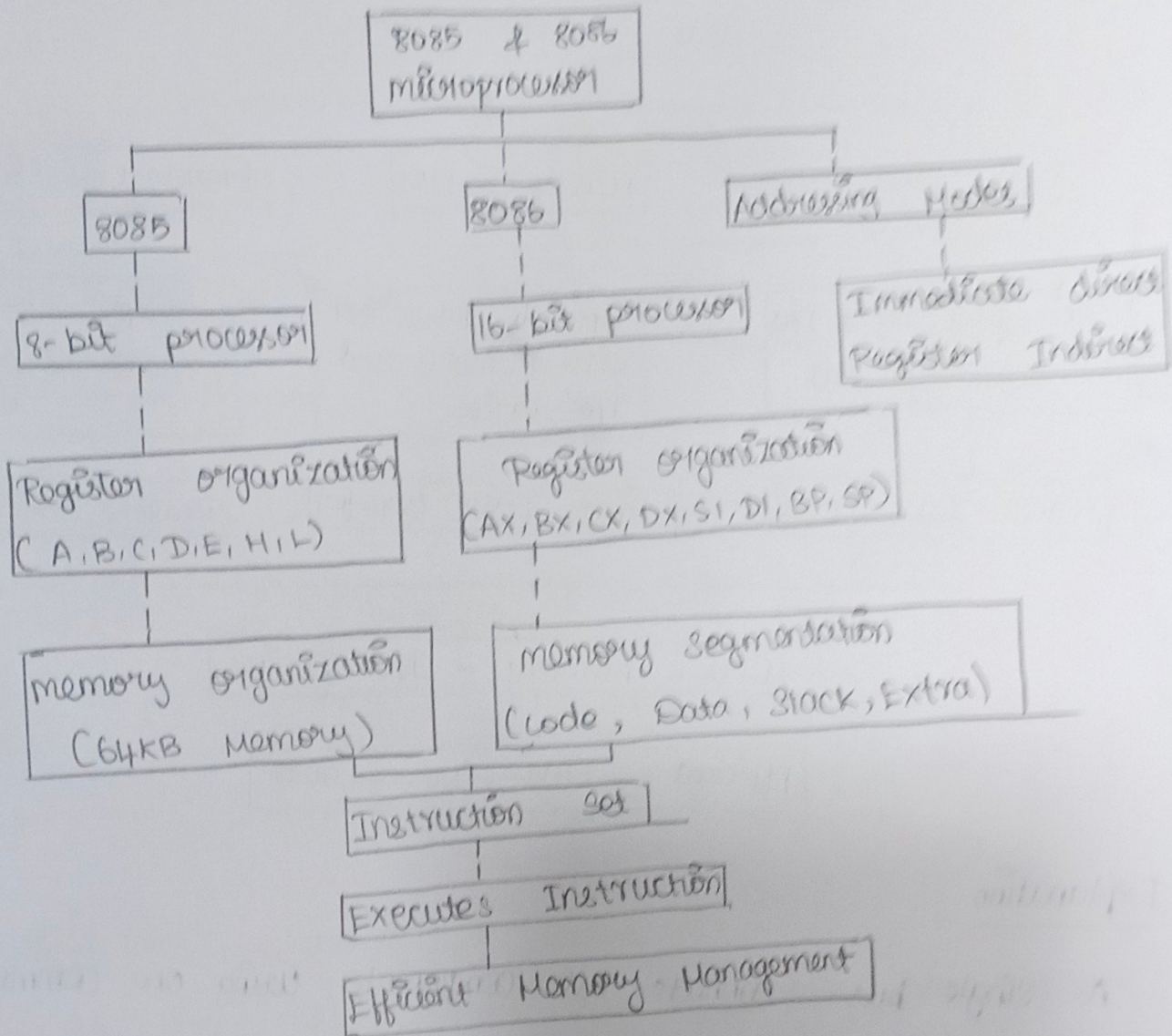


Explanation

A single bus connects all components using one communication path, which may cause delays. A Multi-Bus System uses separate buses for CPU, memory and I/O devices to improve data transfer. A Hierarchical Bus uses multiple bus levels to reduce traffic and improve communication efficiency.

concept map of 8085 and 8086 microprocessor Architecture

concept map



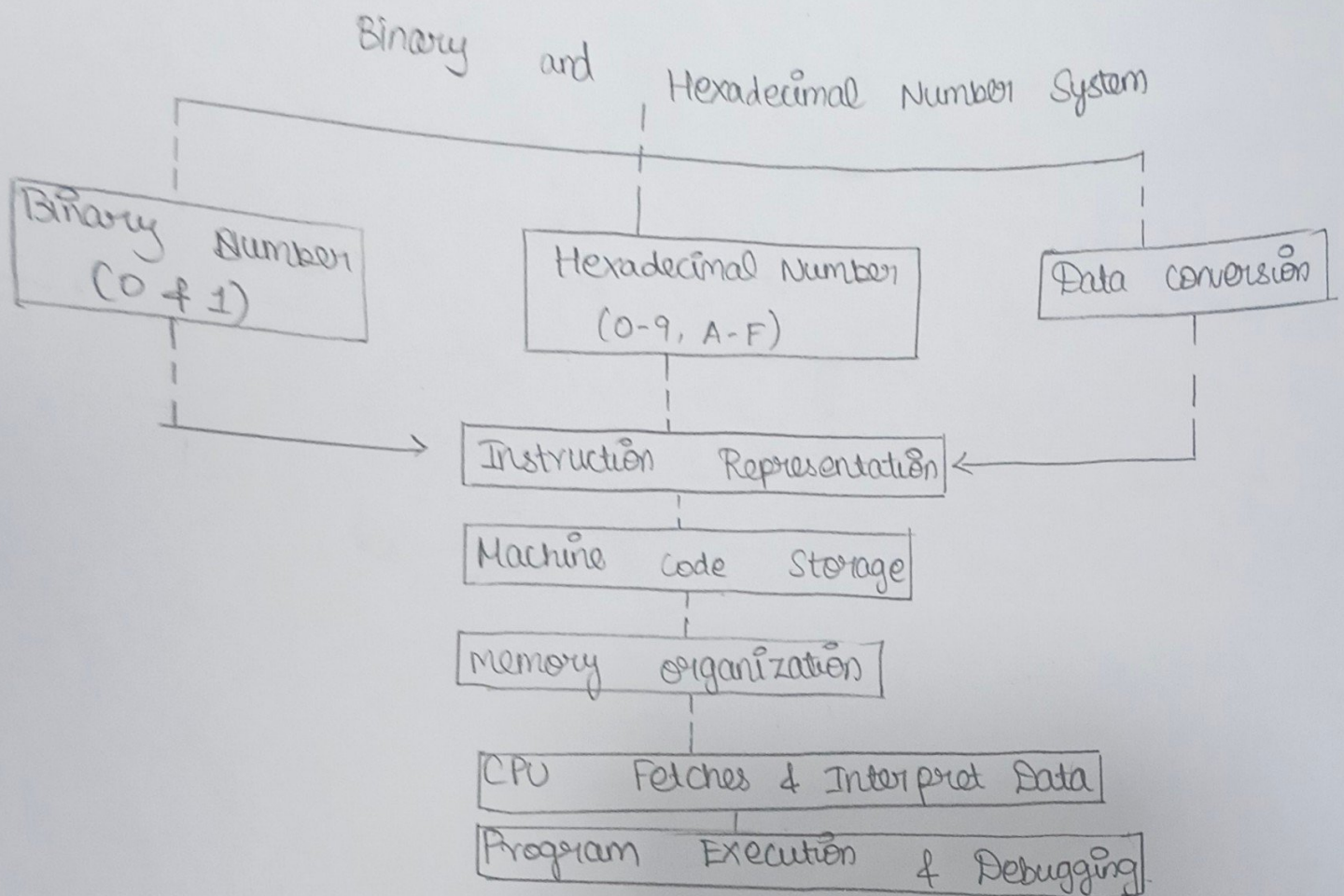
Explanation

The 8085 is an 8-bit microprocessor with 64KB memory and simple register organization. The 8086 is a 16-bit microprocessor with memory segmentation and advanced register. Different addressing modes help the processor access data efficiently.

Structure

concept map of Binary and Hexadecimal Number System

concept map:



Explanation

The Binary Number is used by CPU to process instruction while the Hexadecimal Number System provides a compact representation of Binary values. Data conversion helps represent machine code efficiently, making memory organization and debugging easier.